Zhaohui "Vito" Liang, Ph.D.

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Madison, WI 53719

Education

- **University of Wisconsin - Madison** *Aug 2025*
  PhD, Civil and Environmental Engineering Madison, US
  Advisor: Dr. Xiaopeng(Shaw) Li
- **University of South Florida** *Jan 2020 - Aug 2022*
  PhD Candidate, Civil and Environmental Engineering Tampa, US
  Advisor: Dr. Xiaopeng(Shaw) Li
- **University of Harbin Institute - Weihai** *Jun 2019*
  BS, Civil and Environmental Engineering Shan Dong, China

Research Interests

- Eco-driving control algorithms for energy-efficient and sustainable transportation
- Resilience evaluation of vehicle controllers under real-world disturbances
- Connected and Automated Vehicles (CAVs) as mobile sensing platforms for traffic state estimation and infrastructure evaluation
- Integration of smart infrastructure with CAVs for cooperative mobility and traffic signal optimization
- Telematics and big data analytics for intelligent transportation systems (ITS)
- Transportation Safety

Professional Experience

- **Post Doctoral Research Associate** *Aug 2025 - Present*
  Traffic Operations and Safety Laboratory at University of Wisconsin - Madison
- **Graduate Research Assistance** *Jan 2020 - Aug 2025*
  Connected and Automated Transportation System Laboratory at University of Wisconsin - Madison
- **Lab Manager** *Jan 2022 - Aug 2025*
  Connected and Automated Transportation System Laboratory at University of Wisconsin - Madison

Research Experience

- **[Evaluating energy and mobility impacts of CAV - SMART 2.0]** *Oct 2022 - June 2024*
  Sponsor: Argonne National Lab, $223,000 [ ]
  - **Proposal writing**
  - Collaborated with Argonne National Laboratory to evaluate energy and mobility impacts of connected and automated vehicles (CAVs) under the U.S. DOE SMART 2.0 initiative
  - Developed and calibrated vehicle data acquisition systems integrating CAN bus signals, GNSS/IMU measurements, and V2X communication hardware.
  - Assessed and improved V2X communication performance under varying scenarios and communication technologies (PC5 vs. LTE-Uu)
  - Integrated eco-driving control algorithms with real-time vehicle control interfaces and conducted closed-track experiments to validate system performance.
  - Supported joint analysis and publication efforts on energy efficiency, trajectory smoothing, and system-level CAV benefits
- **[Identifying Cybersecurity Risks in Connected Vehicle Infrastructures]** *May 2020 - Nov 2021*
  Sponsor: Florida Department of Transportation, $152,990
  - Investigated potential cyber attack vectors targeting connected vehicle environments
  - Developed simulation models of cyber attacks on On-Board Units (OBUs) and Roadside Units (RSUs)
  - Demonstrated the impact of various cyber threats on connected vehicle communication and control
  - Proposed mitigation strategies to enhance system-level cybersecurity resilience

- **[Cooperative Automation Research: CARMA Proof-of-Concept TSMO Use Case Testing]** *Jan 2021 - March 2022*
  Sponsor: USDOT FHWA, $189,636
  - Deployed and integrated the FHWA CARMA platform on a research vehicle, including onboard computing, vehicle control interfaces, and ROS-based software components
  - Performed system configuration, hardware troubleshooting, and field testing to support cooperative automated driving applications.
  - Contributed to platform enhancement and supported the development of the TSMO (Transportation Systems Management and Operations) program
- **[CARMA – Integrated Highway Prototype (IHP) Implementation]** *March 2021 - May 2022*
  Sponsor: USDOT FHWA, $75,786 [▶]
  - Deployed and integrated the FHWA CARMA platform on the laboratory test vehicle
  - Performed field testing at Suntrax to evaluate cooperative driving behaviors in highway scenarios
  - Supported the development and refinement of the IHP program through platform enhancements and scenario integration
- **[Infrastructure Enablers for Reliable CDA Phase I: Existing Infrastructure Evaluation]** *Oct 2022 - Jan 2024*
  Sponsor: Florida Department of Transportation, $60,000
  - Conducted a comprehensive literature review and field testing to evaluate the capabilities and limitations of current AV technologies available in the market
  - Identified edge cases and operational challenges in real-world driving environments
  - Provided actionable recommendations for infrastructure enhancements to support reliable and scalable automated traffic systems
- **[Visual-Enhanced Cooperative Traffic Operations (VECTOR) System]** *Oct 2021 - Jan 2025*
  Sponsor: Department of Energy, $850,000 [GitHub]
  - **Proposal writing**
  - Collaborated with industrial partner Connected Wise LLC to the development and validation of the VECTOR system for scalable, low-cost Cooperative Driving Automation (CDA)
  - Supported efforts to reduce infrastructure and system-level costs in sensing, communication, and computation by up to 40%
  - Integrated visual sensing with machine learning to enhance situational awareness for CAV coordination
  - Participated in hardware-software co-design and system integration for real-world deployment
- **[Roadway friction screening and measurement with automated vehicle]** *May 2024 - Aug 2025*
  Sponsor: CCAT, $100,000
  - **Proposal writing**
  - Investigated the use of automated vehicles as mobile probes to monitor and assess roadway surface friction in real time
  - Developed an optimal control-based strategy to induce controlled high-slip maneuvers for enhanced road condition detection
  - Applied tire dynamics modeling to relate vehicle responses to surface friction characteristics under varying conditions
  - Aimed to support infrastructure maintenance and safety by enabling scalable, vehicle-based friction screening methods
- **[NSF I-Corps: Road-bot]** *Mar 2024 - Jun 2024*
  Sponsor: NSF $50,000
  - **Proposal writing**
  - Served as Technical Team Lead, leading platform development
  - Conducted 20+ stakeholder interviews across industry, academia, and government agencies
  - Developed the business model and identified key customer pain points to inform next-stage startup development
- **[Tribal & Rural Autonomous Vehicles for Equity, Livability and Safety (TRAVELS)]** *Jan 2026*
  Sponsor: USDOT $15,000,000

◦ **Proposal writing**

[Severe Weather Connected Vehicle Warning System] *September 2025*

Sponsor: Wisconsin DOT $250,000

• Developed a in-vehicle sensing data collection pipeline

• Developed a computer vision based visibility classification method

JOURNAL ARTICLES

[1] **Liang, Z.**, Ma, C., Zhou, H., Long, K., & Li, X. (2025). **An analytical eco-driving trajectory planning method with field test validation at intersections**. *Transportation Research Part D: Transport and Environment*, *146*, 104870.

[2] Long, K., **Liang, Z.**, Shi, H., Shi, L., Chen, S., & Li, X. (2024). **Traffic oscillation mitigation with physics-enhanced residual learning (PERL)-based predictive control**. *Communications in Transportation Research*, pp. 100154. DOI: 10.1016/j.commtr.2024.100154

[3] Li, Q., Li, X., Yao, H., **Liang, Z.**, & Xie, W. (2023). **Automated vehicle identification based on car-following data with machine learning**. *IEEE Transactions on Intelligent Transportation Systems*, pp. 13893–13902. DOI: 10.1109/TITS.2023.3247985

[4] Shi, X., Yao, H., **Liang, Z.**, & Li, X. (2022). **An empirical study on fuel consumption of commercial automated vehicles**. *Transportation Research Part D: Transport and Environment*, pp. 103253. DOI: 10.1016/j.trd.2022.103253

[5] Long, K., Shi, H., Chen, Z., **Liang, Z.**, Li, X., & de Souza, F. (2024). **Bi-scale car-following model calibration based on corridor-level trajectory**. *Transportation Research Part E: Logistics and Transportation Review*, pp. 103497. DOI: 10.1016/j.tre.2024.103497

[6] Ma, K., Zhou, H., **Liang, Z.**, & Li, X. (2025). **Automated vehicle microscopic energy consumption study (AV-Micro): Data collection and model development**. *Energy*, pp. 135096. DOI: 10.1016/j.energy.2025.135096 (Impact Factor-9.4)

[7] Zhou, H., Zhang, P., **Liang, Z.**, Li, H., & Li, X. (2025). **Coverage trajectory planning problem on 3D terrains with safety constraints for automated lawn mower: Exact and heuristic approaches**. *Robotics and Autonomous Systems*, pp. 105109. DOI: 10.1016/j.robot.2025.105109 (Impact Factor-5.2)

[8] Zhou, H., Ma, C., Shen, S., **Liang, Z.**, & Li, X. (2025). **Towards Full-Scenario Safety Evaluation of Automated Vehicles: A Volume-Based Method.** *Transportation Research Part C: Emerging Technologies, 183, 105485.*.

[9] Liang, Z. (2025). **Towards Robust Eco-Driving: Platform, Algorithms, Scenarios, and Testing** *(Doctoral dissertation, The University of Wisconsin-Madison).*

[10] Ma, C., Li, H., Long, K., Zhou, H., **Liang, Z.**, Li, P., ... & Li, X. (2024). **Real-Time Identification of Cooperative Perception Necessity in Road Traffic Scenarios**. Manuscript available at *SSRN*. DOI: 10.2139/ssrn.4973353. Accepted for Transportation Research Part C: Emerging Technologies.

UNDER REVISION

[11] **Liang, Z.**, Ma, C., Long, K., Zhou, H., & Li, X. (2026). **Robustness and Resilience Evaluation of Eco-Driving Strategies at Signalized Intersections** . Manuscript available at *arXiv*. DOI: 10.48550/arXiv.2601.13389. Submitted to Transportation Research Part D, under review.

[12] Ma, K., **Liang, Z.**, Zhou, H., & Li, X. (2024). **An Advanced Microscopic Energy Consumption Model for Automated Vehicle: Development, Calibration, Verification**. Manuscript available at *arXiv*. DOI: 10.48550/arXiv.2408.11797

[13] Li, X., **Liang, Z.**, Shi, X., & Yao, H. (2021). **Vehicle-based Sensing for Energy and Emission Reduction**. Manuscript available at *arXiv*.

[14] Ma, K., Zhang, Y., Zhou, H., **Liang, Z.**, & Li, X. (2024). **Real-world Automated Vehicle Longitudinal Stability Analysis: Controller Design and Field Test**. Manuscript available at *SSRN*. DOI: 10.2139/ssrn.4960777

Conference Papers

[1] **Liang, Z.**, Han, J., Li, X., Karbowski, D., Ma, C., & Rousseau, A. (2024). **Testing Cellular Vehicle-to-Everything Communication Performance and Feasibility in Automated Vehicles**. *In 2024 IEEE Intelligent Vehicles Symposium (IV)*, pp. 2917–2922. DOI: 10.1109/IV54607.2024.10333876

[2] **Liang, Z.**, Ma, C., Zhou, H., Long, K., & Li, X. (2025) . **Towards Robust Eco-Driving: An Adversarial Framework for Resilience Evaluation Under Realistic Disturbances**. *Accepted for 2026 TRB Annual Meeting Oral Presentation.*

[3] **Liang, Z.,** Ma, C., Zhou, H., Long, K., & Li, X. (2025) . **Robustness and Resilience Evaluation of Eco-Driving Strategies at Signalized Intersections**. *Accepted for 2026 TRB Annual Meeting Oral Presentation.*

[4] **Liang, Z.**, Zhou, H.,Huang, H.& Li, X. (2026). **High-Slip-Ratio Control for Peak Tire-Road Friction Estimation Using Automated Vehicles** . *Accepted for 2026 IEEE Intelligent Vehicles Symposium (IV)*

[5] Li, H., Cao, B., **Liang, Z.**, Ma, C., Zhou, H.,& Li, X. (2025) . **CATS-V2V: A Real-World Vehicle-to-Vehicle Cooperative Perception Dataset with Complex Adverse Traffic Scenarios**. *Accepted for ASCE's 2026 Transportation Conference*

[6] Ma, K., Zhang, Y., Zhou, H.,**Liang, Z.**, & Li, X. (2025).**Real-world Automated Vehicle Longitudinal Stability Analysis: Controller Design and Field Test**. *In 2025 IEEE International Conference on Robotics and Automation (ICRA) (pp. 1-7). IEEE.*

[7] Li, H., Cao, B., **Liang, Z.**, Li, W., Oh, J., Chen, Y., Liang, S., Zhou, H., Ma, C., Liu, J. & Li, Z. (2025). **CATS-V2V: A Real-World Vehicle-to-Vehicle Cooperative Perception Dataset with Complex Adverse Traffic Scenarios.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2026.

Patents

[1] Li, X., Li, Q., Yao, H., & **Liang, Z.** (2023). **Automated vehicle identification based on car-following data with machine learning** *U.S. Patent Application No. 17/884,411*.

[2] Ma, K., Li, X., Zhang, P., & **Liang, Z.** (2024). **A New Vehicle-to-Vehicle Communication Technology Based on Binary Light Code**. *University of South Florida, Tampa, FL (United States)*.

[3] Ma, K., Huang, H., Li, H., Zhang, P., **Liang, Z.**, & Li, X. (2025). **A New Vehicle-to-Vehicle Communication System: Visual-Enhanced Cooperative Traffic Operations**. *University of Wisconsin–Madison*.

Skills

- **Vehicle Hardware & System Integration:** Drive-by-Wire (DBW), CAN Bus, LiDAR, Radar, GNSS/IMU, Onboard Computing, V2X OBU/RSU, Sensor Calibration, Hardware Troubleshooting
- **Autonomous Driving Systems:** ROS/ROS2, Autoware, Real-Time Vehicle Control, Hardware-in-the-Loop (HIL) Testing
- **Programming Languages:** C++, C, Python, Matlab
- **Data Science & Machine Learning:** TensorFlow, PyTorch, pandas, NumPy
- **DevOps & Version Control:** Git, Docker, Overleaf
- **Specialized Area:** Robot Operating System (ROS) ,Vehicle-to-Everything (V2X), CAN Bus Protocol, Simulink, Embedded system, Multi-thread programming
- **Mathematical & Statistical Tools:** CasADi, Matlab, ipopt
- **Other Tools & Technologies:** LaTeX, Linux/Ubuntu, Vissim, CARLA Simulator, IPG CarMaker

Teaching and Leadership Experience

- **Research Mentor(Digital Credential)** *November 2026*
  University of Wisconsin–Madison, Graduate School Delta Research Mentor Training Program
- **Teaching Support** *Spring 2026*
  Civil And Environmental Engr 570 - Connected and Automated Transportation System
  - Help prepare slides and lab material
- **Guest lecturer** *Dec 2024*
  Civil And Environmental Engr 574 - Traffic Control
  - Delivered a guest lecture on CV2X technology and its cybersecurity vulnerabilities in connected vehicle systems
- **Guest lecturer** *Feb 2026*
  Civil And Environmental Engr 570 - Connected and Automated Transportation System
  - Delivered a guest lecture

- **Research Mentor** *Sep 2024 - May 2025*
  Electrical And Computer Engr 399 - Independent Study
  - Supervised undergraduate students on connected and automated vehicle research projects
  - Organized and led monthly research meetings to guide progress and provide technical support

- **Invited Talk** *Oct 31, 2025*
  University of Wisconsin-Milwaukee transportation seminar Fall 2025
  - Eco-driving Control Algorithm

- **Invited Talk** *Jan 11, 2026*
  Transportation Research Board Annual Meeting 2026
  - Towards Robust Eco-Driving: An Adversarial Framework for Resilience Evaluation Under Realistic Disturbances
  - Robustness and Resilience Evaluation of Eco-Driving Strategies at Signalized Intersections

- **Invited Talk** *March 24, 2022*
  7th Annual UTC Conference
  - From Theoretical Models to Real-World Implementations: The USF Connected Automated Vehicle Testbed

- **President** *Sep 2021 - Sep 2022*
  Institute of Transportation Engineers (ITE) Student Chapter, University of South Florida
  - Organized social events, technical workshops, and outreach activities for student members
  - Coordinated field visits and networking opportunities with transportation professionals
  - Led the organization of the student Traffic Bowl competition team and related activities

Outreach

CAV Lab Tour Host, Wisconsin Department of Transportation Bring-Your-Kids-To-Work Day, UW-Madison, 7/15/2024
CAV Lab Tour Host, Oshkosh Corp. Visit Program, UW-Madison, 5/10/2024
CAV Lab Tour Host, Ariens Corp. Visit Program, UW-Madison, 4/11/2024
CAV Lab Tour Host, BECOME Visit Program, UW-Madison, 3/31/2023
CAV Lab Tour Host, Soloway Felix Visit Program, UW-Madison, 4/02/2024
AV Ride Demonstration Host, 2025 AAA Safe Mobility Conference, Madison, WI, 4/7/2025
AV Ride Demonstration Host, WisDOT Summer of Safety Open House- Hill Farms, Madison, WI, 5/19/2025
AV Ride Demonstration Host, WCHA 2025 Summer Highway Conference, Dells, WI, 6/2/2025
CAV Lab Tour Host, WisDOT BTO UW TOPS Lab Tour, UW-Madison, 11/18/2025
AV Demonstration Host, Florida Automated Vehicles Summit, 12/14/2022
AV Demonstration Host, 7th Annual UTC Conference, 3/24/2022

Professional Memberships

- **SAE International**, Student Member *Mar 2023 - Present*
- **IEEE**, Student Member *Jun 2025 - Present*

Additional Information

Languages: Chinese (Native speaker), English (Proficient)
Interests: Robotics, Racing, Basketball, Badminton, Swimming!